1. prior belief abt the robot-
   - where is it? state : x
   - How sure I am? certainty : P

2. Control inputs Arrive
   - what im trying to do now
   - Forward/Side speeds, turn rate
     → V_x, V_y, W_z → over dt

3. Predict motion in the world using the changes in x, y, θ

4. How errors propagate as we move

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$$S = \underbrace{C_t \cdot P_t \cdot C_t^T}_{\text{Error covariance}} + Q_t$$

S → Residual Covariance (expected combined uncertainty of prediction and measurment)

C_t → Observation matrix (describes how to map the state x_t to an observation z_t) (Kxn)

C_t^T → transpose

Q_t → measurment-noise covariance (uncertainty in measurments)

P_t → the uncertainty in prediction + measurment

$$K_t = P_t \cdot C_t^T \cdot S^{-1}$$

S^{-1} → Residual Covariance

K_t → Kalman gain (Balanced factor for blending prediction and measurments)

$$y = \boxed{z_t} - \boxed{C_t \; x_t}$$

z_t → Real measurments observation Input- measurment-residual (innovation) the difference between real measurment- and prediction)

C_t → Observation Matrix

x_t → current- position → gives the predicted state observation

z_t → Input- from GPS measurments (x,y)

$$x_t = x_t + K_t \cdot y$$

K_t → Kalman Gain

y → measurment-residual

correcting x based on observation

$$P = (I - K_t C_t) \cdot P$$

I → Identity matrix

C_t → observation matrix

K_t → Kalman Gain

updating error

matrix multiplication → @ input-

prediction ($F @ x_{t+1} + B @ (u)) = x$

$P @ (P_{t-1} @ F.T) + Q = P$

(transpose)

correction

$S = H @ (P @ H.T) + R$

$K = (P @ H.T) @ S^{-1}$

$y = z - (H @ x)$ → input-

$x = x + (K @ y)$

$I =$

$P = (I - (K @ H)) @ P$

Covariance tells us if variables are positively or inversly related.

Transpose : Rows become columns

$A \cdot A^T = (A)^2$

prev. location + location change through the controller

current state

$X_t = A_t \cdot X_{t-1} + B_t u_t$ → input- from cmd-vel (v, w)

(3×3) (3×1) (3×2) (2×1)

prev. state

(Jacobian) system model (state transition matrix)
↳ How the state evolves from t-1 to t without control or noise
$(n \times n)$

(control matrix (effect of control input-)
↳ describes how the control u_t changes the state from t-1 to t
$(n \times 1)$

Basically the formulas that gets the inputs and move the robot-

prediction

Error Covariance

$P_t = A_t \cdot P_{t-1} \cdot A_t^T + R_t$

(3×3) (3×3) (3×3)

prev. Error Covariance

transpose of A

↳ process noise (uncertainty in the process)

updating covariance

$P' = APA^T$ A = matrix A^T = transpose of A P = covariance matrix

↓

how uncertain each variable in X is and how they vary with each other

- IF we have a random vector x with covariance P:

  new variable → $x' = Ax$

  then the covariance of x' is given by: $P' = APA^T$

motion model in the code: $x_t = \begin{bmatrix} x + V_x\, dt \cos(\theta) \\ y + V_x\, dt \sin(\theta) \\ \theta + w\, dt \end{bmatrix}$ x direction / y direction / Rotation

taking partial derivative wrt x, y, θ $= \begin{bmatrix} 1 & 0 & -dt\, V_x \sin(\theta) \\ 0 & 1 & dt\, V_x \cos(\theta) \\ 0 & 0 & 1 \end{bmatrix}$ → Jacobian

control input → $u_k = \begin{bmatrix} V_x \\ w \end{bmatrix}$ ↗ velocity in x, y

$Ax_k + Bu_k = x_t \rightarrow \begin{bmatrix} x_k \\ y_k \\ \theta_k \end{bmatrix} + \begin{bmatrix} dt \cos(\theta_k) & 0 \\ dt \sin(\theta_k) & 0 \\ 0 & dt \end{bmatrix} \begin{bmatrix} V_x \\ w \end{bmatrix}$ (B matrix, control)

↳ Rotation in z

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